

Solution Requirements

Date	27 June 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements(FR)** define specific behaviors, Functions or features or the system(What system should do).
- **Non-Functional Requirements(NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors(How the system should be).

Functional Requirements:

Category	Description	Priority
Authentication	Users can securely register, log in, and access the platform using JWT-based authentication and encrypted passwords. High	High
Authorization	Only authenticated borrowers can access their own financial profile, loan records, settlement recommendations, and AI-generated negotiation history.	High
External Interfaces	The system integrates with Google Gemini AI API for negotiation strategy generation and uses FastAPI APIs for frontend-backend communication.	High

Transaction Processing	The platform processes loan information, calculates financial health metrics, predicts settlement amounts, and generates negotiation letters.	High
Reporting	Displays borrower dashboard with financial health score, EMI ratio, debt stress level, settlement recommendations, AI history, and reports.	Medium
Business Rules	Settlement recommendations are generated based on income, expenses, overdue period, loan amount, and financial stress level.	High
Compliance	User information is securely stored using password hashing and JWT authentication while maintaining borrower data privacy.	High
Other	Stores AI-generated negotiation letters, settlement history, and financial analysis for future reference.	Medium

Non-Functional Requirements

Category	Requirement	Target
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Performance	APIs should return financial analysis and dashboard data quickly.	Response time below 2 seconds
Scalability	The application should support increasing borrowers and loan records without performance degradation.	Supports multiple concurrent users
Security	Secure user authentication using JWT, password hashing, and protected APIs.	Encrypted passwords and secure authentication
Reliability	The platform should remain stable during financial calculations and AI processing.	99.9% uptime
Usability	Responsive, user-friendly interface for desktop and mobile devices.	Desktop/Mobile
Maintainability	Modular architecture for easy maintenance and future feature enhancements.	Easy updates
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